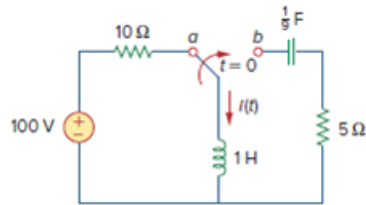


Problem

The circuit in Fig. 8.12 has reached steady state at $t = 0^-$. If the makebefore-break switch moves to position b at $t = 0$, calculate $i(t)$ for $t > 0$.

Figure 8.12:



Step-by-step solution

Step 1 of 7

Draw the circuit diagram:

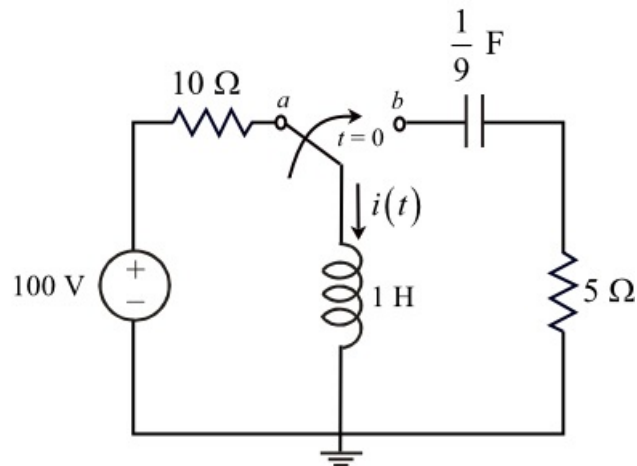


Figure 1

Step 2 of 7

At $t = 0^-$ the switch is connected to position 'a'. The capacitor charges to the supplied dc voltage. The equivalent circuit is shown in Figure 2.

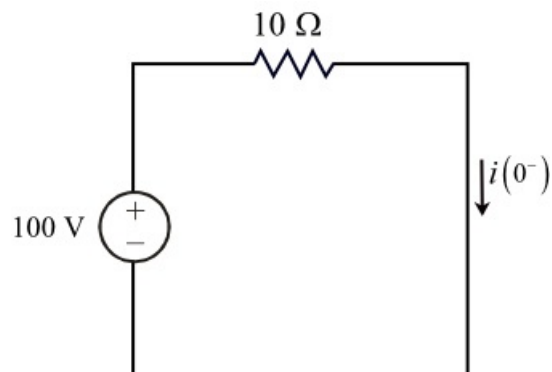


Figure 2

Step 3 of 7

Calculate the initial current using Ohm's law.

$$i(0^-) = \frac{100}{10} \\ = 10 \text{ A}$$

Initial voltage across capacitor is $v(0) = 0 \text{ V}$

At $t = 0^+$ the switch is connected to position 'b'. Draw the modified circuit diagram.

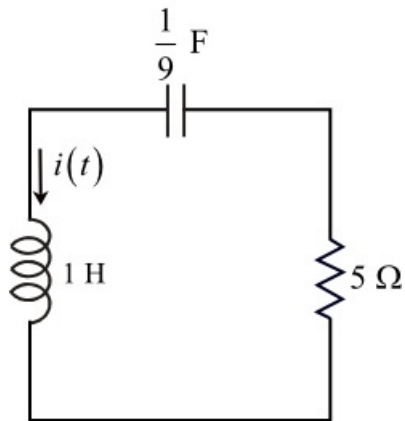


Figure 3

Apply Kirchhoff's voltage law to the Figure 1.

$$Ri(t) + L \frac{di(t)}{dt} + v(t) = 0$$

Substitute 0 for t in the equation $Ri(t) + L \frac{di(t)}{dt} + v(t) = 0$.

$$Ri(0) + L \frac{di(0)}{dt} + v(0) = 0$$

$$\begin{aligned} \frac{di(0)}{dt} &= \frac{-1}{L} (Ri(0) + v(0)) \\ &= \frac{-1}{1} ((5)(10) + 0) \\ &= -50 \text{ A/s} \end{aligned}$$

Therefore, the initial value of $\frac{di(0)}{dt}$ the 50 A/s.

Calculate the neper frequency.

$$\begin{aligned} \alpha &= \frac{R}{2L} \\ &= \frac{5}{(2)(1)} \\ &= 2.5 \end{aligned}$$

Calculate the resonant frequency.

$$\begin{aligned} \omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{1 \times \frac{1}{9}}} \\ &= 3 \end{aligned}$$

[Comments \(1\)](#)



Anonymous

what happen to negative sign of rate of change of current

Step 5 of 7

Calculate the roots of characteristic equation.

$$\begin{aligned} s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \\ &= -2.5 \pm \sqrt{2.5^2 - 9} \\ &= -2.5 \pm j1.6583 \end{aligned}$$

Since $\alpha < \omega$, the response is under damped.

Therefore, the response is

$$i(t) = e^{-2.5t} (A_1 \cos 1.6583t + A_2 \sin 1.6583t) \text{ A} \dots (1)$$

Substitute 0 for t in equation (1).

$$\begin{aligned} i(0) &= e^{-2.5(0)} (A_1 \cos 1.6583(0) + A_2 \sin 1.6583(0)) \text{ A} \\ 10 &= A_1 \\ A_1 &= 10 \end{aligned}$$

Step 6 of 7

Differentiate the equation (1) with respect to 't'.

$$\frac{di(t)}{dt} = \left[-2.5e^{-2.5t} (A_1 \cos 1.6583t + A_2 \sin 1.6583t) + e^{-2.5t} (1.6583A_1 (-\sin 1.6583t) + 1.6583A_2 (\cos 1.6583t)) \right]$$

$$\left. \frac{di(t)}{dt} \right|_{t=0} = 1(0 + A_2 1.6583) - 2.5(A_1 + 0)$$

$$\frac{di(0)}{dt} = -2.5A_1 + 1.6583A_2$$

$$\frac{di(0)}{dt} = -2.5A_1 + 1.6583A_2$$

$$-2.5A_1 + 1.6583A_2 = -50 \dots\dots (2)$$

Step 7 of 7

Substitute 10 for A_1 in equation (2).

$$(-2.5)(10) + (1.6583)(A_2) = -50$$

$$1.6583A_2 = -50 + 25$$

$$A_2 = \frac{-25}{1.6583}$$

$$= -15.0756$$

Substitute the values of A_1 and A_2 in equation (1).

$$i(t) = e^{-2.5t} (10 \cos 1.6583t - 15.0756 \sin 1.6583t) \text{ A}$$

Hence the current $i(t)$ is $e^{-2.5t} (10 \cos 1.6583t - 15.0756 \sin 1.6583t) \text{ A}$.